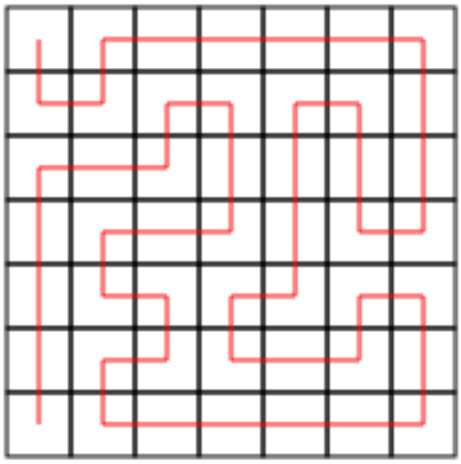
Grid Path Description

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1625 | Introductory Problems | https://cses.fi/problemset/task/1625

There are 88418 paths in a 7 × 7 grid from the upper-left square to the lower-left square. Each path corresponds to a 48-character description consisting of characters D (down), U (up), L (left) and R (right).

For example, the path corresponds to the description DRURRRRRDDDLUULDDDLDRRURDDLLLLLURULURRUULDLLDDDD.



You are given a description of a path which may also contain characters ? (any direction). Your task is to calculate the number of paths that match the description.

Input

The only input line has a 48-character string of characters ?, D, U, L and R.

Output

Print one integer: the total number of paths.

Example

Input
??????R??????U???????????????????????????LD????D?

Output
201